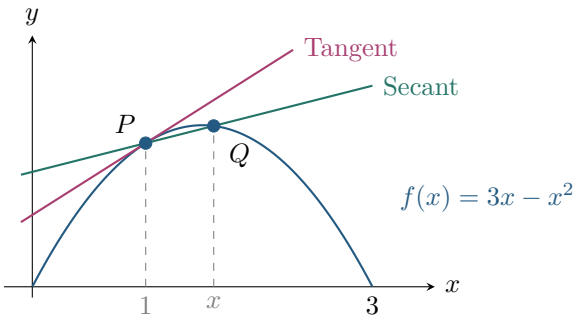
Derivatives

MATH 1334 October 12, 2026

Instantaneous velocity and tangent lines

Newton's question. How do we compute instantaneous velocity from position?

Suppose a ball's position is $f(x) = 3x - x^2$. What is its velocity at $x = 1$? We need the slope of the tangent line. Approximate it using secant lines.



The secant slope is the *difference quotient*:

$$\frac{\Delta y}{\Delta x} = \frac{f(x) - f(1)}{x - 1}.$$

Taking its limit gives the tangent slope:

$$\begin{aligned}\lim_{x\to 1} \frac{f(x) - f(1)}{x - 1} &= \lim_{x\to 1} \frac{3x - x^2 - 2}{x - 1} \\ &= \lim_{x\to 1} \frac{-(x-2)(x-1)}{x - 1} \\ &= \lim_{x\to 1} -(x - 2) = 1.\end{aligned}$$

Thus the instantaneous velocity at $x = 1$ is 1.

The derivative at a point

Definition. The *derivative of f at a*, read "f prime at a," is

$$f'(a) = \lim_{x\to a} \frac{f(x) - f(a)}{x - a} = \lim_{h\to 0} \frac{f(a + h) - f(a)}{h},$$

provided the limit exists as a finite number. The two forms agree by setting $h = x - a$. We say f is *differentiable at a* when $f'(a)$ exists.

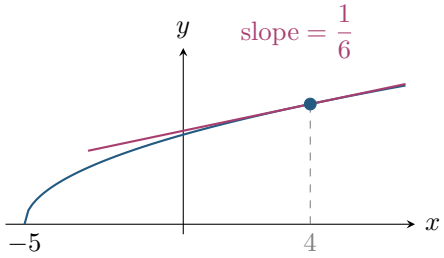
The number $f'(a)$ represents the slope of the tangent line to f at $x = a$, or the instantaneous rate of change of f there. The tangent line is the best linear approximation near that point.

Computing derivatives from the definition

Example 1. For $f(x) = \sqrt{x+5}$, find $f'(4)$. *Exam 1: 2b*

Conjugate. Since $f(4) = 3$,

$$\begin{aligned} f'(4) &= \lim_{x\to 4} \frac{\sqrt{x+5}-3}{x-4} \\ &= \lim_{x\to 4} \frac{(\sqrt{x+5}-3)(\sqrt{x+5}+3)}{(x-4)(\sqrt{x+5}+3)} \\ &= \lim_{x\to 4} \frac{x+5-9}{(x-4)(\sqrt{x+5}+3)} \\ &= \lim_{x\to 4} \frac{1}{\sqrt{x+5}+3} = \frac{1}{6}. \end{aligned}$$



Example 2. For $f(x) = |x|$, find $f'(0)$.

Case method.

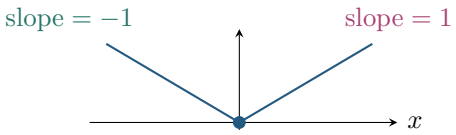
$$f'(0) = \lim_{h\to 0} \frac{|0+h|-|0|}{h} = \lim_{h\to 0} \frac{|h|}{h},$$

if this limit exists. Check each side:

$$L^- = \lim_{h\to 0^-} \frac{-h}{h} = -1,$$

$$L^+ = \lim_{h\to 0^+} \frac{h}{h} = 1.$$

Since $L^- \neq L^+$, the limit does not exist. Therefore $f'(0)$ does not exist.



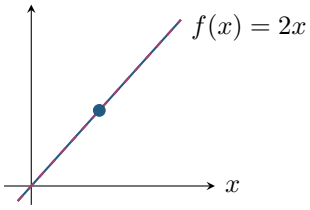
There is no single tangent slope at the corner.

Example 3. For $f(x) = mx + b$, find $f'(2)$.

Difference quotient.

$$\begin{aligned} f'(2) &= \lim_{x\to 2} \frac{f(x) - f(2)}{x - 2} \\ &= \lim_{x\to 2} \frac{(mx + b) - (2m + b)}{x - 2} \\ &= \lim_{x\to 2} \frac{m(x - 2)}{x - 2} = m. \end{aligned}$$

The choice $a = 2$ does not matter: $f'(a) = m$ for every a. A line is its own tangent line.



Section 3.2: the derivative function

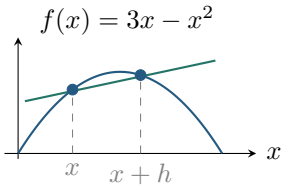
Definition. For a function f, the *derivative function* f' is defined by

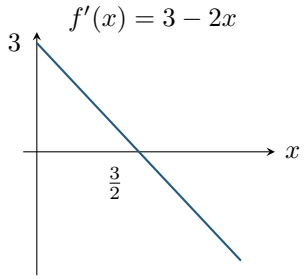
$$f'(x) = \lim_{h\to 0} \frac{f(x + h) - f(x)}{h}.$$

Its domain consists of the points x where this limit exists as a finite number.

Example. Let $f(x) = 3x - x^2$. Then

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{[3(x + h) - (x + h)^2] - [3x - x^2]}{h} \\ &= \lim_{h\to 0} \frac{3x + 3h - x^2 - 2xh - h^2 - 3x + x^2}{h} \\ &= \lim_{h\to 0} \frac{h(3 - 2x - h)}{h} \\ &= \lim_{h\to 0} (3 - 2x - h) = 3 - 2x. \end{aligned}$$



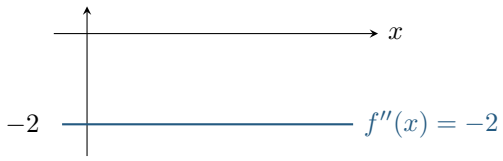


Higher-order derivatives

Continue with $f(x) = 3x - x^2$ and $f'(x) = 3 - 2x$.

Second derivative. The derivative of f' is f'', read "f double prime."

$$\begin{aligned} f''(x) &= \lim_{h\to 0} \frac{f'(x+h) - f'(x)}{h} \\ &= \lim_{h\to 0} \frac{[3 - 2(x+h)] - [3 - 2x]}{h} \\ &= \lim_{h\to 0} \frac{-2h}{h} = -2. \end{aligned}$$



Third derivative. The derivative of f'' is f''', read "f triple prime."

$$\begin{aligned} f'''(x) &= \lim_{h\to 0} \frac{f''(x+h) - f''(x)}{h} \\ &= \lim_{h\to 0} \frac{-2 - (-2)}{h} = \lim_{h\to 0} \frac{0}{h} = 0. \end{aligned}$$

The fourth derivative is also zero. In fact,

$$f^{(n)}(x) = 0 \qquad \text{for every integer } n \geq 3.$$

Notation

1. Prime/Newton notation. For the nth derivative, write

$$f^{(n)}(x) = f^{\overbrace{\prime\cdots\prime}^{n \text{ primes}}}(x).$$

In particular, $f^{(2)} = f''$ and $f^{(3)} = f'''$.

2. Leibniz notation. Write

$$\frac{df}{dx} = f'(x).$$

At $x = a$,

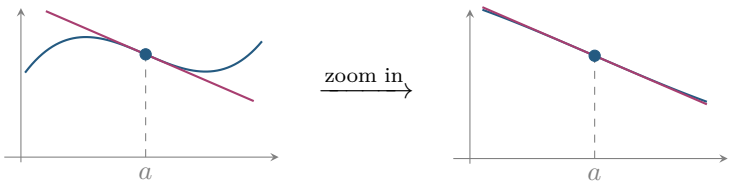
$$\frac{df}{dx}(a) = \left.\frac{df}{dx}\right|_{x=a} = f'(a).$$

The symbol $\dfrac{d}{dx}$ is the *derivative operator*.

Differentiability implies continuity

Theorem. *Let a be in the domain of f. If $f'(a)$ exists, then f is continuous at a.*

Geometric idea. When we zoom in near a differentiable point, the graph is approximated by its tangent line, which is continuous at a.



Proof. We want to show $\lim_{x\to a} f(x) = f(a)$. For $x \neq a$,

$$f(x) = (x-a)\frac{f(x)-f(a)}{x-a} + f(a).$$

Since the difference quotient has the finite limit $f'(a)$, the limit laws give

$$\begin{aligned}\lim_{x\to a} f(x) &= \Big(\lim_{x\to a}(x-a)\Big)\left(\lim_{x\to a}\frac{f(x)-f(a)}{x-a}\right) + f(a)\\ &= 0\cdot f'(a) + f(a) = f(a).\end{aligned}$$

Therefore f is continuous at a. □

Corollary. *If $f'(a)$ exists, then the tangent line at $(a, f(a))$ is*

$$y - f(a) = f'(a)(x-a), \qquad \textit{or} \qquad y = f'(a)(x-a) + f(a).$$

Proof. Use point-slope form with slope $f'(a)$ and point $(a, f(a))$. □

The converse is false

Continuity does not imply differentiability.

Example. The function $f(x) = |x|$ is continuous at 0, since $\lim_{x\to 0} |x| = 0 = f(0)$. Yet $f'(0)$ does not exist:

$$\lim_{h\to 0^-}\frac{|h|}{h} = -1 \qquad \text{and} \qquad \lim_{h\to 0^+}\frac{|h|}{h} = 1.$$

Practice

1. Find the tangent line to $f(x) = \sqrt{x+6}$ at $x = 3$.

Conjugate. Since $f(3) = 3$,

$$\begin{aligned} f'(3) &= \lim_{x\to 3} \frac{\sqrt{x+6}-3}{x-3} \\ &= \lim_{x\to 3} \frac{(\sqrt{x+6}-3)(\sqrt{x+6}+3)}{(x-3)(\sqrt{x+6}+3)} \\ &= \lim_{x\to 3} \frac{x-3}{(x-3)(\sqrt{x+6}+3)} \\ &= \lim_{x\to 3} \frac{1}{\sqrt{x+6}+3} = \frac{1}{6}. \end{aligned}$$

Thus the tangent line is

$$y - 3 = \frac{1}{6}(x-3), \qquad \text{or} \qquad y = \frac{x}{6} + \frac{5}{2}.$$

2. Identify f and a in the limit below, and find $f'(a)$:

$$\lim_{h\to 0} \frac{(1+h)^2 - 1}{h}.$$

Match the definition. Take $f(x) = x^2$ and $a = 1$. Then $f(1+h) = (1+h)^2$ and $f(1) = 1$, so

$$f'(1) = \lim_{h\to 0} \frac{1+2h+h^2-1}{h} = \lim_{h\to 0}(2+h) = 2.$$

3. Is f differentiable at $x = -1$? Prove your answer rigorously.

$$f(x) = \begin{cases} (x+1)^2, & x \neq -1, \\ 1, & x = -1. \end{cases}$$

(a) Use continuity. We have $\lim_{x\to -1} f(x) = 0$, but $f(-1) = 1$. Thus f is discontinuous at -1 and therefore is not differentiable there.

(b) Use the definition. For $x \neq -1$,

$$\frac{f(x) - f(-1)}{x+1} = \frac{(x+1)^2 - 1}{x+1} = (x+1) - \frac{1}{x+1}.$$

The one-sided limits of this quotient are

$$\lim_{x\to -1^-} \left((x+1) - \frac{1}{x+1}\right) = +\infty, \qquad \lim_{x\to -1^+} \left((x+1) - \frac{1}{x+1}\right) = -\infty.$$

Hence $f'(-1)$ does not exist.